Ajinkya Jain

Senior Research Scientist, Google (Intrinsic) | Ph.D., UT Austin

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Research Profile & Interests

Senior Research Scientist at **Google (Intrinsic)** with **5+ years of post-PhD R&D experience** advancing generalist robot manipulation, **on-policy distillation and reinforcement finetuning**, **LLM/VLM policy synthesis**, and **force-aware imitation and online learning** for contact-rich tasks. Author of **14 peer-reviewed papers and preprints** across CoRL, ICRA, IROS, RSS Workshops, IEEE RA-L, and IEEE T-RL (**3,700+ citations**; **ICRA 2024 Best Conference Paper Award**), co-inventor of a granted US robotics patent, research mentor to **9 PhD interns and graduate students**, and co-organizer of **3 flagship workshops** at RSS, IROS, and IEEE Humanoids (2026).

Research Interests: Robot Learning, Embodied Foundation Models (VLMs/LLMs), Reinforcement Learning & Policy Distillation, Contact-Rich & Whole-Body Robot Control, Sim-to-Real Transfer, Manipulation & Motion Planning Under Uncertainty

Education

The University of Texas at Austin — Ph.D., Mechanical Engineering (*Robotics & Controls*) 2015–2021

Advisors: Prof. Scott Niekum (Computer Science) & Prof. Ashish Deshpande

Indian Institute of Technology Kanpur — B.Tech.–M.Tech., Mechanical Engineering 2010–2015

Dual Degree in Robotics and Automation | Advisor: Prof. Bishakh Bhattacharya

Industry Research Experience

Senior Research Scientist, Google (Intrinsic), Mountain View, CA Nov 2026–Present

Senior Software Engineer (Research), Google (Intrinsic), Mountain View, CA Oct 2023–Oct 2026

Robotics Engineer III (Research), Google (Intrinsic), Mountain View, CA Nov 2022–Sep 2023

R&D Team (Lead: Prof. Stefan Schaal) — Promoted twice following Alphabet's acquisition of Vicarious

- **RL Finetuning & Multi-Robot Policy Distillation:** Co-developed *RoboBallet2* (under review, *IEEE T-RL*), scaling collision-free multi-robot reaching across arbitrary 3D geometries via on-policy distillation and reinforcement finetuning in collaboration with Google DeepMind and UC San Diego.
- **Foundation Models & Compliant Policy Synthesis:** Co-designed *GenCHiP* (*IROS 2024*), equipping LLMs with compliant, force-constrained action spaces to synthesize robot policy code for tight-tolerance, contact-rich manipulation.
- **Force-Aware Imitation & Online Adaptation:** Formulated efficient online contact-force model learning (*ICRA 2025*) and predictive force-conditioned imitation learning (*RSS 2026 Workshop*) for sub-mm industrial connector insertion under uncertainty.
- **Multimodal & Cross-Embodiment Robot Learning:** Co-authored cross-lab research on sketch-conditioned imitation learning (*RT-Sketch*, *CoRL 2024*), bimanual imitation learning benchmarks (*IEEE RA-L 2024*), and cross-embodiment datasets and RT-X foundation models (*Open X-Embodiment*, *ICRA 2024 Best Conference Paper Award*).
- **Industrial Tech Transfer & Patented IP:** Architected generalizable manipulation, perception, and object placement pipelines (**US Patent 12,330,304 B2**) deployed across **10+ high-fidelity rigid-body assembly and insertion benchmarks**.
- **Research Leadership & Mentorship:** Scoped research agendas and mentored **6 PhD research interns and 3 Master's students** (Stanford, CMU, UCSD, USC, and Arizona State University collaboration), leading directly to multiple co-authored publications.

Robotics Engineer III, Vicarious FPC, Inc. (Acquired by Alphabet, Union City, CA) Oct 2021–Nov 2022

- Developed constrained object manipulation, grasp planning, and tight-fit insertion algorithms for robotic assembly and warehouse automation.
- Implemented motion planning pipelines minimizing planning and trajectory execution cycle times while achieving $\geq 95\%$ task success rates.

- Devised data-driven 3D perception and pose estimation algorithms for high-precision object placement, establishing core IP transitioned into Google (Intrinsic) (**US Patent 12,330,304 B2**).

AI Research Intern, Samsung AI Research Center (Robotics), New York, NY May 2019–Aug 2019

- Developed *GrAB-Net*, a fully autonomous RGB-D imitation learning pipeline generating category-level optimal approach and grasping behaviors from single-view depth observations ($\sim$ 70% grasp success across 40 unseen objects).

Robotics Research Intern, Vicarious FPC, Inc., Union City, CA May 2018–Aug 2018

- Implemented platform- and task-agnostic visual servoing interfaces in Python for high-precision ($\leq$ 2 mm) robotic assembly with fast target convergence ($\leq$ 20 s) and high repeatability ($\geq$ 90% success).

Academic Research Experience

Graduate Research Assistant, Dept. of Computer Science, UT Austin, Austin, TX Aug 2015–Aug 2021

- **Category-Independent Articulation Perception (*CoRL 2021, ICRA 2021*):** Invented *ScrewNet* and distributional depth-based screw-theory architectures to estimate 3D object articulation models directly from raw depth images without prior category labels (**2× data efficiency** over category-specific baselines).
- **Learning Hybrid Object Kinematics from Observations (*IROS 2020*):** Developed a changepoint-based method to learn planning-compatible hybrid kinematics models for articulated objects from human demonstrations (**up to 3× higher accuracy** over state-of-the-art methods).
- **Manipulation Motion Planning Under Uncertainty (*CoRL 2018, RSS 2017 Workshop*):** Formulated a trajectory-optimization POMDP motion planner leveraging contact and object interactions for long-horizon manipulation (**5× faster planning**, $\geq$ 90% task success).

Graduate Research Assistant, Dept. of Mechanical Engineering, IIT Kanpur, India May 2014–May 2015

- **Master's Thesis (*Exoskeleton Gripper Optimization & Bipedal Gait Control*):** Formulated a computationally lightweight analytical model for piezoelectric stack actuators (*Springer AISC 2014*), optimized robotic gripper design via multi-objective NSGA-II (*Structural and Multidisciplinary Optimization 2015*), and designed a time-varying LTV-LQR controller for bipedal exoskeleton locomotion.

Undergraduate Visiting Researcher, Texas A&M University, College Station, TX May 2013–Aug 2013

- **Belief-Space Motion Planning (*Advisor: Prof. Suman Chakravorty*):** Modeled dynamics across 5 holonomic and non-holonomic mobile robot platforms, implemented Feedback Information RoadMaps (FIRM) for navigation under uncertainty, and built a synchronized real-time control interface with V-REP.

Publications & Patents [Google Scholar: 3,700+ Citations]

Manuscripts Under Review

[1] Y. Zhi, M. Lai, **A. Jain**, G. Salhotra, K. Go, G. Ustensvik, T. Kroeger, S. Schaal, and M. C. Yip, "RoboBallet2: Multi-robot Reaching for Arbitrary Geometries via On-policy Distillation and Reinforcement Finetuning," *Under Review at IEEE Transactions on Robot Learning (T-RL)*, 2026.

Refereed Journal & Conference Publications

[1] K. Tracy, Z. Manchester, **A. Jain**, K. Go, S. Schaal, T. Erez, and Y. Tassa, "Efficient Online Learning of Contact Force Models for Connector Insertion," *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[2] P. Sundaresan, Q. Vuong, J. Gu, P. Xu, T. Xiao, S. Kirmani, T. Yu, M. Stark, **A. Jain**, K. Hausman, D. Sadigh, J. Bohg, and S. Schaal, "RT-Sketch: Goal-Conditioned Imitation Learning from Hand-Drawn Sketches," *Conference on Robot Learning (CoRL)*, 2024.

[3] K. Burns, **A. Jain**, K. Go, F. Xia, M. Stark, S. Schaal, and K. Hausman, "GenCHiP: Generating Robot Policy Code for High-Precision and Contact-Rich Manipulation Tasks," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

[4] Open X-Embodiment Collaboration (incl. **A. Jain**), "Open X-Embodiment: Robotic Learning Datasets and RT-X Models," *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. **[Best Conference Paper Award]**

[5] M. Drolet, S. Stepputtis, S. Kailas, **A. Jain**, J. Peters, S. Schaal, and H. Ben Amor, "A Comparison of Imitation Learning Algorithms for Bimanual Manipulation," *IEEE Robotics and Automation Letters (RA-L)*, 2024.

[6] **A. Jain**, S. Giguere, R. Lioutikov, and S. Niekum, "Distributional Depth-Based Estimation of Object Articulation Models," *Conference on Robot Learning (CoRL)*, 2021.

[7] **A. Jain**, R. Lioutikov, C. Chuck, and S. Niekum, "ScrewNet: Category-Independent Articulation Model Estimation From Depth Images Using Screw Theory," *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[8] **A. Jain** and S. Niekum, "Learning Hybrid Object Kinematics for Efficient Hierarchical Planning Under Uncertainty," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020.

[9] **A. Jain** and S. Niekum, "Efficient Hierarchical Robot Motion Planning Under Uncertainty and Hybrid Dynamics," *Conference on Robot Learning (CoRL)*, 2018.

[10] R. Datta, **A. Jain**, and B. Bhattacharya, "A Piezoelectric Model Based Multi-Objective Optimization of Robot Gripper Design," *Structural and Multidisciplinary Optimization*, Springer, 2015.

[11] **A. Jain**, R. Datta, and B. Bhattacharya, "Unified Minimalistic Modelling of Piezoelectric Stack Actuators for Engineering Applications," *Advances in Intelligent Systems and Computing*, Springer, 2014.

Refereed Workshop Publications

[1] Y. Korkmaz, K. Go, G. Salhotra, E. Biyik, **A. Jain**, and S. Schaal, "Predictive Force Conditioning for Contact-Rich Imitation Learning," *Workshop on RoboData, Robotics: Science and Systems (RSS)*, 2026.

[2] **A. Jain** and S. Niekum, "Belief Space Planning under Approximate Hybrid Dynamics," *Workshop on POMDPs in Robotics, Robotics: Science and Systems (RSS)*, 2017.

Granted Patents

[1] A. Sriraman, **A. Jain**, and J. C. Wendelken, "Object Placement," *US Patent 12,330,304 B2*, Granted June 2025 (Google / Intrinsic).

Invited Talks & Presentations

- "AI-Enabled Robots: Towards Real-World Applications," *Karlsruhe Institute of Technology (KIT)* Nov 2024
- "Learning and Leveraging Task Dynamics for Robot Manipulation Under Uncertainty," *Dexterity, Inc.* 2021
- "Learning Hybrid Object Kinematics for Efficient Hierarchical Planning Under Uncertainty," *IEEE/RSJ IROS* 2020
- "Learning Hybrid Object Kinematics from Human Demonstrations," *UT Austin Robotics Seminar* 2020
- "Learning and Leveraging Object Kinematics for Robot Manipulation Under Uncertainty," *SAIC-New York* 2019
- "Efficient Hierarchical Robot Motion Planning Under Uncertainty and Hybrid Dynamics," *CoRL* 2018

Professional Service & Community Leadership

Workshop Organizer

- "Frontiers in Robust Whole-Body Control: Methods, Systems, and Real-World Deployment," *IEEE Humanoids* 2026
- "Bridging Learning and Physics for Robust Whole-Body Robot Control," *IEEE/RSJ IROS* 2026
- "Closing the Sim2Real Gap: Progress and Challenges," *Robotics: Science and Systems (RSS)* 2026

Editorial & Peer Review Service

- **Conference Session Chair:** IEEE International Conference on Robotics and Automation (*ICRA*) 2021
- **Journal Reviewer:** *IEEE Transactions on Robotics (T-RO)*, *IEEE Robotics and Automation Letters (RA-L)*, *Mechatronics*
- **Conference Reviewer:**
  - **Robotics: Science and Systems (RSS):** 2024, 2025, 2026
  - **Conference on Robot Learning (CoRL):** 2021, 2022, 2023, 2024, 2025, 2026
  - **IEEE Int. Conf. on Robotics and Automation (ICRA):** 2019, 2020, 2021, 2023, 2025, 2027
  - **IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS):** 2020, 2021, 2022, 2025, 2026
  - **IEEE-RAS Int. Conf. on Humanoid Robots (Humanoids):** 2016 | **IEEE ARSO:** 2017

Mentorship & Teaching Experience

- **Research Mentor**, Google (Intrinsic), Mountain View, CA 2022–Present
  Mentored 6 PhD/graduate research interns and 3 Master's students (Arizona State University collaboration) across robot RL finetuning, online contact-force learning, and bimanual/force-conditioned imitation learning
- **Guest Lecturer**, Dept. of Mechanical Engineering, UT Austin Spring 2016
  Mechatronics (ME 340) — Lecture on Python Programming for Robotics & Mechatronics
- **Graduate Teaching Assistant**, Dept. of Mechanical Engineering, UT Austin 2015–2016
  Dynamic Systems and Controls Laboratory (ME 144L)
- **Graduate Teaching Assistant**, Dept. of Mechanical Engineering, IIT Kanpur 2014–2015
  Optimization Methods in Engineering Design (ME 752) & Mechanical Engineering Laboratory II (ME 471N)

Honors, Awards & Scholarships

- **Best Conference Paper Award**, IEEE Int. Conf. on Robotics and Automation (*ICRA*) 2024
  Open X-Embodiment: Robotic Learning Datasets and RT-X Models
- **Second Runner-up Worldwide**, RoboCup@Home SSPL League (Team Austin Villa), Nagoya, Japan 2017
- **AI Laboratory Travel Grant**, The University of Texas at Austin 2017, 2018
- **TAMU–IITK Student Exchange Research Scholarship**, Texas A&M University 2013
- **Boeing Corporation Robotics Scholarship**, Indian Institute of Technology Kanpur 2012–2013
- **Certificate of Merit for Academic Excellence**, Indian Institute of Technology Kanpur 2011–2013
- **CBSE Merit Scholarship (AIEEE)** for Undergraduate Engineering Studies 2010–2014
- **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship**, DST, Govt. of India 2009

Technical Expertise

- **Robot Learning & Embodied AI:** Reinforcement Learning & RL Finetuning, On-Policy Distillation, Imitation Learning / Behavior Cloning, Foundation Models & LLMs/VLMs for Robotics, Policy Code Synthesis, Cross-Embodiment Datasets, Sim-to-Real Transfer
- **Control, Planning & Perception:** Compliant Impedance & Whole-Body/Force-Torque Control, Online Contact-Force Modeling, Trajectory Optimization (SNOPT, IPOPT), POMDP / Belief-Space Planning, Multi-Robot Coordination, 3D Depth & Pose Perception, Visual Servoing (ViSP)
- **Frameworks & Simulation:** Python, C++, MATLAB/R, PyTorch, JAX, TensorFlow, MuJoCo, ROS, PyBullet, Gazebo, CoppeliaSim/V-REP, OMPL, MoveIt!, GraspIt!, PCL, OpenCV, Kubernetes
- **Robot Hardware:** KUKA Industrial Arms, Universal Robots (UR), Kinova (Jaco2 6/7-DOF, Gen3), Bimanual Manipulation Setups, Toyota Human Support Robot (HSR), Rethink Sawyer, Robotiq Grippers, 6-Axis Force-Torque Sensors